

PROJECT DESIGN PHASE-II : TECHNOLOGY STACK

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

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Developer ID: 23JR1A05E4 (Individual)

Developer Name: Shaik Zuveriya Tabassum

Max Marks: 4 Marks

1. TECHNICAL ARCHITECTURE

The deliverable includes the high-level architecture demarcation guidelines, component mappings, and the integrated multi-tier SaaS platform structure.

Architecture Demarcation Guidelines:

- **Processes & Application Logic:** Catalog dynamic form rendering, dynamic matrix fare engine, QR payload generator, and ticket logging.
- **Infrastructural Demarcation:** ServiceNow High-Availability Enterprise Cloud Multi-Instance Architecture (SaaS).
- **Data Storage & External Interfaces:** Custom ServiceNow tables (x_sz_metro_ticket) with external REST APIs for SMS / OAuth integration.

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[CLIENT LAYER]
  Service Portal UI (Mobile & Web Browser) | Station Manager UI (QR Scanner App)
    | HTTPS / REST
[SERVICENOW NOW PLATFORM (APPLICATION LOGIC)]
  Service Catalog Engine → Client Scripts & UI Policies
  Script Includes / Fare Logic Matrix Distance Calculator
  QR Payload Generator (Base64 / JS QR Library)
  REST Notification API (Outbound Email / SMS)
    | JDBC / MariaDB
[DATA & CLOUD LAYER]
  ServiceNow Cloud DB (x_sz_metro_ticket table)
  External Gateways (Twilio SMS / OAuth SSO)
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2. COMPONENT & TECHNOLOGY STACK DETAILS (TABLE-1)

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	User Interface	How user interacts with application e.g. ServiceNow Service Portal, Mobile Catalog, UI Pages	HTML5, CSS3, JavaScript, AngularJS, Bootstrap 3
2	Application Logic-1	Service Catalog Form Validation, UI Policies, and Client-side dynamic field dependency logic	ServiceNow Catalog Client Scripts & UI Policies
3	Application Logic-2	Backend dynamic fare computation engine using matrix station lookup scripts	Server-side JavaScript (ServiceNow Script Includes / Business Rules)
4	Application Logic-3	Dynamic QR Code Generation Payload & UI Page rendering engine	JavaScript QRCode.js Library, ServiceNow UI Pages
5	Database	Custom application table definitions storing ticket transaction logs, stations, and fare rules	ServiceNow Relational Schema (MariaDB / MySQL engine)
6	Cloud Database	Enterprise ServiceNow Cloud managed database instance with automatic failover and replication	ServiceNow Enterprise Cloud DB (x_sz_metro_ticket)
7	File Storage	Storage for station map assets, generated PDF tickets, and system attachment logs	ServiceNow Attachment API / Cloud DB Storage
8	External API-1	SMS notification gateway API for sending ticket booking confirmation payloads to passenger mobile numbers	Twilio REST API / ServiceNow Outbound REST
9	External API-2	User identity authentication and Single Sign-On (SSO) gateway integration	OAuth 2.0 / Google Gmail OAuth / SAML 2.0
10	Machine Learning	Predictive passenger traffic analysis and rush-hour volume forecasting (Optional / Analytics)	ServiceNow Predictive Intelligence / Machine Learning Engine
11	Infrastructure	Application deployment and hosting on cloud platform with automated backups	ServiceNow Cloud Platform (Multi-tenant SaaS Architecture)

3. APPLICATION CHARACTERISTICS (TABLE-2)

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	Open-Source Frameworks	Client-side portal rendering, responsive layouts, and QR generation libraries	AngularJS, Bootstrap, QRCode.js (Open-Source JS)
2	Security Implementations	Role-Based Access Control (RBAC), Encrypted QR payloads, and HTTPS data encryption in transit	SHA-256 Payload Encryption, ServiceNow ACLs, TLS 1.3, IAM Controls
3	Scalable Architecture	Decoupled 3-tier architecture separating Presentation Portal, Application Engine, and Storage Layer	ServiceNow Platform 3-Tier SaaS Architecture
4	Availability	High-availability cloud environment utilizing automated failover data centers and load balancers	ServiceNow Advanced High Availability (AHA) Infrastructure
5	Performance	Optimized client-side rendering, server-side caching of fare tables, and minimal query response times (<1.5s)	ServiceNow In-Memory Caching, CDN asset delivery

References: <https://c4model.com/> | <https://developer.servicenow.com/> | <https://www.ibm.com/cloud/architecture> | <https://aws.amazon.com/architecture>